Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 13)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Table 1 shows some features of Earth, Mars and Venus.

Table 1

Feature	Venus	Earth	Mars
Mean distance from Sun / million km	108	150	228
Mass relative to Earth	0.82	1.0	0.11
Atmospheric pressure relative to Earth	92	1.0	0.006
Mean surface temperature / °C	460	15	-60
Magnetic field	Very weak	Strong	Very weak
Rotation period / hours	5 800	24	25

0 1 . 1 Use Table 1 to explain why liquid water cannot exist on the surface of Mars.

[3 marks]

0 1 . 2 Explain how the Earth's mass and its distance from the Sun created suitable conditions for life.

[3 marks]

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0 1 . 3 Explain the importance of the Earth's magnetic field and its speed of rotation for living organisms.

[2 marks]

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0 1 . 4 Explain how the tilt of the Earth's axis affects living organisms.

[2 marks]

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0 2 A student sampled the ground-flora of two woodlands. Table 1 shows the results for woodland A.

Table 1

Species	Number of individuals (n)
Bluebell	40
Wood anemone	30
Dog's mercury	20
Bramble	10
Total (N)	100

0 2 . 1 Use the formula $D = N(N - 1) / \Sigma n(n - 1)$ to calculate Simpson's index of diversity for woodland A.

[3 marks]

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Answer

0 2 . 2 Woodland B had an index of 1.6. State what this shows and give **one** limitation of using the index alone to compare the woodlands.

[2 marks]

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0 3 . 1 Explain how each of the following environmental impacts of salmon farming arises and how it can be reduced.

[8 marks]

Overfishing of wild fish for feed

Sea lice and disease

Escapes affecting wild gene pools

Organic waste pollution

0 3 . 2 Explain why culling seals and cormorants near fish farms is controversial and give an alternative.

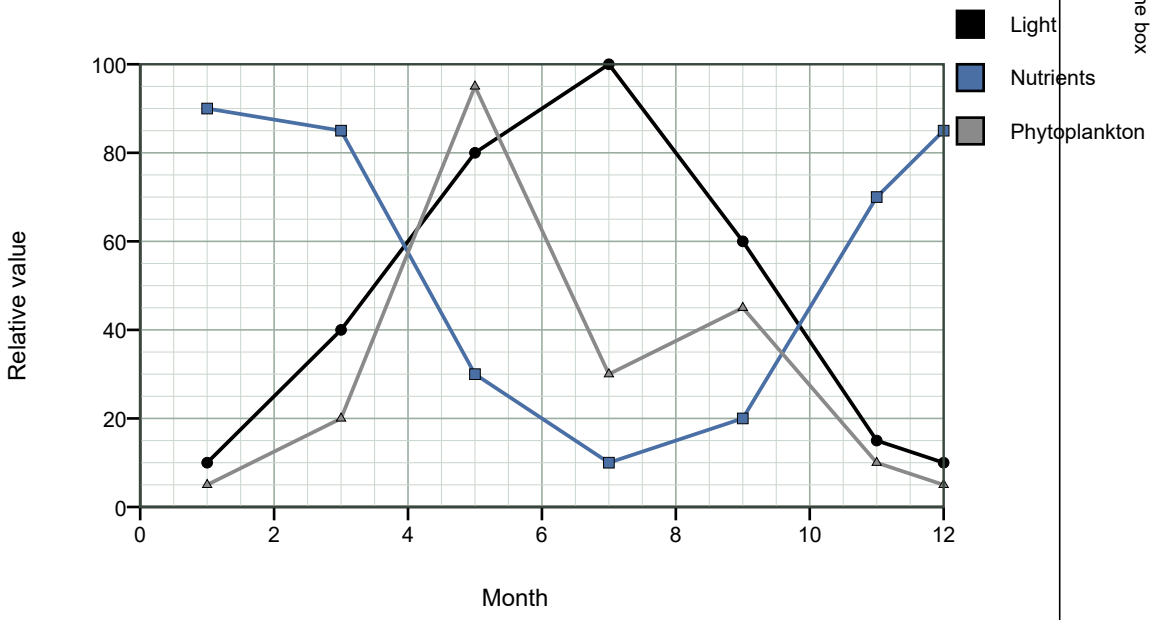
[2 marks]

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0 4 Figure 1 shows the seasonal changes in light intensity, nutrient concentration and phytoplankton biomass in the surface water of a temperate ocean in the northern hemisphere.

Figure 1



0 4 . 1 Use Figure 1 to explain the spring peak and summer decline in phytoplankton biomass.

[4 marks]

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0 4 . 2 Explain why tropical open oceans have low productivity all year despite high light and temperature.

[2 marks]

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0 4 . 3 Explain why areas of upwelling and coastal waters are the most productive parts of the sea.

[3 marks]

0 4 . 4 Explain why phosphate is often the limiting nutrient in the sea.

[1 mark]

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0 5 . 1 Explain how the relationship between coral polyps and zooxanthellae allows reefs to support a high biomass in nutrient-poor tropical seas.

[4 marks]

0 5 . 2 Describe the abiotic conditions needed by reef-building corals and explain why these make reefs vulnerable.

[3 marks]

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0 5 . 3 Explain **three** ways in which coral reefs are important to humans.

[3 marks]

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0 6 Table 1 lists some feedback mechanisms that affect the Earth's climate.

0 6 . 1 Complete Table 1 by stating whether each mechanism is an example of **negative** or **positive** feedback.

[4 marks]

Table 1

Mechanism	Type of feedback	
Higher temperatures increase cloud cover, raising albedo		
Melting permafrost releases methane		
Higher CO_2 levels increase photosynthesis and carbon sequestration		
Melting ice reduces albedo		

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0 6 . 2 Explain why natural systems regulated by negative feedback rarely reach tipping points.

[2 marks]

0 6 . 3 Using an example, explain how a large unusual event can cause a tipping point.

[2 marks]

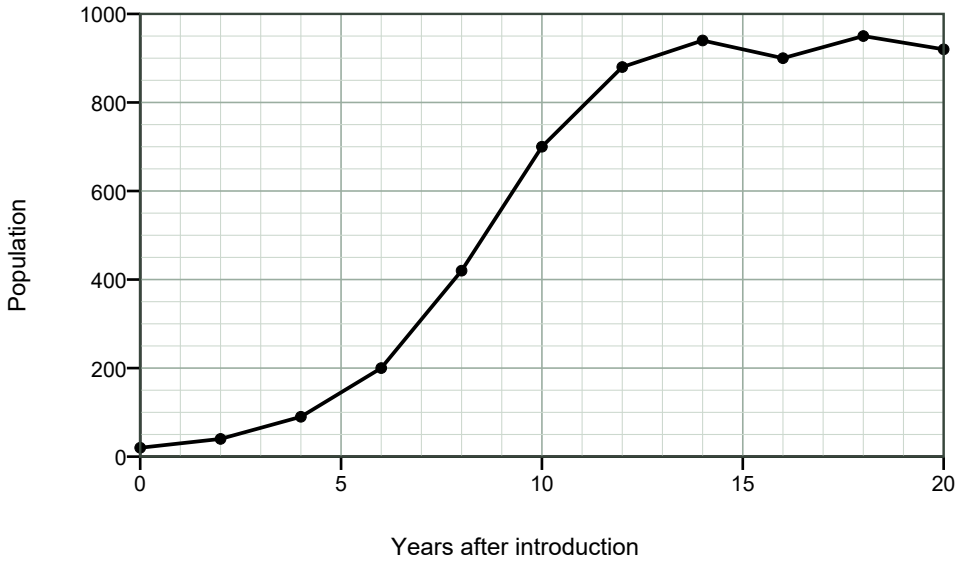
0 6 . 4 Explain how the human response to negative feedback in agriculture can make food production less sustainable.

[2 marks]

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0 7 Figure 1 shows the population of red deer on an island after they were introduced in year 0. No predators are present.

Figure 1



0 7 . 1 Name the shape of the growth curve and explain the changes in growth rate shown.

[4 marks]

0 7 . 2 Explain the difference between density-dependent and density-independent factors, with an example of each.

[2 marks]

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0 7 . 3 Explain why the deer population on the island may need to be culled even though it appears stable.

[2 marks]

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0 7 . 4 Explain how predator and prey populations regulate each other.

[2 marks]

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0 8 Table 1 shows typical energy values for an agroecosystem. Values are units of energy remaining from 100 000 units of solar energy.

Table 1

Stage	Energy / arbitrary units
Solar energy reaching the crop	100 000
Energy captured during photosynthesis	4 000
Energy stored in crop biomass	1 000

0 8 . 1 Calculate the percentage of the solar energy reaching the crop that is stored in crop biomass.

[1 mark]

Answer ☐ %

0 8 . 2 Explain why so little of the solar energy is stored in the crop.

[3 marks]

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0 8 . 3 Explain why few food chains have more than four trophic levels.

[2 marks]

0 8 . 4 Explain why rearing sheep or cattle is the best option for food production in upland and semi-arid areas.

[2 marks]

0 8 . 5 Explain how pigs can increase the efficiency of a food production system.

[2 marks]

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0 9 A conservation organisation is converting 700 ha of arable farmland on the coast into a wetland reserve for breeding and wintering birds (as at Wallasea Island).

0 9 . 1 Explain why habitat area and shape are important in the design of the reserve.

[4 marks]

0 9 . 2 Explain how the reserve could be designed to provide a range of niches.

[3 marks]

0 9 . 3 Explain why some species may need to be deliberately introduced while others colonise naturally.

[2 marks]

0 9 . 4 Explain the value of a biological corridor linking the reserve to other wetlands.

[1 mark]

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10 . 1 Discuss the reasons why the conservation of biodiversity is important for human society.

[9 marks]

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Extra space

1 0 . 2 State what is meant by **biodiversity**.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss how ecological monitoring and the development of new technologies have improved the planning and evaluation of conservation strategies.

[25 marks]

OR

1 1 . 2 Discuss the extent to which the principles of the circular economy could be used to develop sustainable lifestyles.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS